Maine Technology Institute Assessment Response

TEAM: Skills and Experience

Our multidisciplinary team combines academic expertise with practical experience across technology, education, and maritime history:

Dr. Scott Eaton - Chair of Engineering Department, USM
Specializes in cross-cutting research focused on improving efficiency and reducing emissions in energy and transportation systems. His work with AI engineering encompasses code development for embedded systems and the collection and analysis of water sensor data throughout Maine.

Dr. Carrie Diaz Eaton - Education and Research Innovation, Computational Mathematics Professor, Bates College
Dedicated to breaking down barriers in math, science, and technology education. Her expertise in pedagogical approaches and curriculum development is essential for making AI instruction accessible to rural Maine communities.

Anthony Scarlatos - CS, Digital Media and Technology, Stony Brook, NY
Brings expertise in interface design, human-computer interaction, mobile application development, and computing for social good. His technical skills are crucial for creating user-friendly applications that serve the public interest.

Rick Sawyer - Bussiness owner and Maritime Historian
Blue Hill resident with extensive knowledge of Maine's maritime heritage and entrepreneurial experience. His research background and local connections provide valuable historical context and community insights.

Steve Brookman - Operations and Community Engagement
Retired United Airlines pilot and organizer of the Blue Hill Heritage Festival. His operational experience and community leadership skills contribute to project management and public outreach efforts.

Bonnie Sparks - Executive Director of Maine University Distance Education - Retired. Thirty years advocating for the non-traditional student in rural Maine.

PROBLEM and VISION: Addressing Critical Gaps

Our initiative addresses two interconnected challenges facing rural Maine:

Problem 1: Limited AI Education Access
Rural Maine communities lack accessible AI instruction opportunities, leaving knowledge workers vulnerable in an increasingly automated economy. This digital divide threatens to exacerbate existing economic disparities.

Problem 2: Inaccessible Historical Archives
Maine's rich maritime history remains locked away in archives, accessible only to researchers with specialized knowledge and significant time investment. This limits public engagement with our cultural heritage.

Vision

As a rural community, we have the opportunity to direct technology instead of being pushed by it. TheAI universe is evolving far too fast, while it brings benefits of personal and industrial efficiencies, it also brings bad actors willing to monetize the technology. There is a plethora of narratives about the future of AI which need to be sorted out individually as true or false, beneficial or harmful. Individuals need to be educated before they can coalesce into the best possible narrative for of our community. This approach, Astoria, offers the common subject of history with a lower entry barrier to understanding the technology behind AI, it is my belief, and my hope that good individual judgment follows.

This proposal envisions a comprehensive solution that democratizes both AI education and historical access, using Maine's maritime heritage as a compelling case study for teaching AI skills while preserving and sharing our cultural legacy.

VALUE PROPOSITION: Unified Solution for Dual Challenges

Our approach uniquely addresses both issues through a single, integrated platform:

1. **Universal Relevance**: Maine maritime history resonates with Maine residents, providing us a common ground.
2. **Technical Excellence**: The Astoria maritime application demonstrates rigorous software engineering principles, economically possible only using AI code generation.
3. **Educational Framework**: The platform serves as both a historical resource and a practical learning environment for AI application development.

PRODUCT: Development and Commercialization Strategy

Product Deliveries:

1: **Astoria is a web-based application** accessible via simple URL and capable responding to complex maritime historical queries, in natural language (NL).

2: **AI Engineering tool for education.** Since the code base, documentation and runtime is available, the training is based in reverse engineering practices, snd using Gemini LLM as the tutor, explaining the parts and the data flow of the application, it should be like watching a transparent car engine while it is running.

3: **Application template** for similar uses. The Hub-Spoke architecture is simple and flexible. The Spokes are services of structured data, vector data and user inquiries. The Hub it's the brain amalgamating all the inputs from the spokes. In a similar scenario, there are thousands of water sensors constantly monitoring sea and fresh water. Sensors output structured data which is stored in many repositories, some are relational databases, excel spreadsheets, etc. There is plenty of environmental text base available in text format. An application based on the Astoria template would provide environmental scientists, a panoramic view of the ever changing conditions that sensors report.

4. **Pedagogy is the study of teaching methods.** Using an AI tutor to teach AI engineering it's not a common breakfast. We have the opportunity to document the pros and cons of this new method even if limited to andragogy.

Open Source Approach: Released under MIT license to ensure:

- Complete transparency with full documentation and source code
- Prevention of corporate data appropriation and monetization of public historical records
- Sustainable community-driven development

Testing and Validation: Prototype development will involve iterative testing with local historical societies, libraries, and educational institutions to ensure usability and accuracy.

MARKET: Competitive Landscape and Positioning

Market Experience: With over four decades in software development since 1980, including positions at Digital Equipment Corporation (DEC), Fidelity, Bank of Boston, IBM, and MetLife, I bring extensive experience across four generations of computing technology. This background enables rapid, efficient code generation across multiple languages including Python, SQL, JSON, and HTML.

Competitive Advantage: By operating as a non-profit initiative, we eliminate the profit motive that drives commercial competitors to restrict access through subscription models. This approach benefits for-profit educational institutions and non-profit organizations alike.

Target Market: Primary focus on historians, adult education programs, and non-profit organizations seeking technical validation and educational resources.

BUSINESS MODEL: Sustainable Non-Profit Framework

My experience in the non-profit arena rests with https://farmsteadartists.org This is an organization my wife Bonnie and I founded in 2020, Farmstead Artists supports independent artist to display and sell their artwork. Our home venue is the Farmstead Barn in Sullivan, Maine. It has become a neighborhood art center - we had no idea this rural area was so rich in talent.

Our last monthly show in July grossed revenue of $8050. The organization is self-directed. My participation in the last year has been almost none. I was able to focus on AI.

The business model adapted was "a large yard sale"; we will require each participant to have a Maine sales tax account. All the activities are conducted by committees. Our mission is "To support each other"

I did think extensively about this business model, I have confidence that it will evolve according to the participants.

SCALE: Growth and Expansion Plans

Current Foundation Model: Our initial dataset includes comprehensive maritime logs, only from the Port of Machias:

- 731 documented vessels
- 2,357 crew member records
- 5,244 voyage entries
- Historical coverage from 1750-1920

Expansion of Context: The infrastructure is designed to accommodate millions of records and can easily scale to include:

- Maritime logs from all Maine ports, and their round-trip voyages.
- Historical documents in multiple formats (PDF, text, and others)
- Narrative content and archival materials statewide

Expansion of Technical Features:

- Image support should enhance responses to queries significantly. Multimodal LLMs and image libraries like LangGraph are now available.

- OpenSource LLM, like Llama should reduce the cost of API usage

- Self-contained solutions should offer data protection and expanded security so you didn't answer OK

- The Hub-Spoke architecture supports expansion of services, data, and user base without significant infrastructure changes.

EXIT/SUSTAINABILITY: Long-Term Success Framework

Mission Completion: This initiative is designed as a catalyst to educate Maine, rather than a permanent institution. Success is measured by the creation of sustainable, collaborative social organizations that can independently maintain and expand AI education in Maine.

Legacy Planning: The ultimate goal is to establish a network of community-based historical and educational organizations equipped with the tools and knowledge to continue this work autonomously.

Knowledge Transfer: The knowledge is AI applications and it is tranfered thruogh education.

I look forward to our discussion and the opportunity to explore how MTI's support can help bring this vision to fruition for the benefit of Maine's communities and the preservation of our maritime heritage.